

Task 6

R code:

```
# 1. Load libraries
library(Hmisc)
library(knitr)
library(vegan)
library(factoextra)
library(plotly)

# 2. Read data - skip header, assign English names
data_raw <- read.table(
  "/Users/apple/Downloads/INRTU courses/2nd semester/Data Analysis/Homework
6/data_morphometry.txt",
  header = TRUE,
  sep = "\t",
  fileEncoding = "CP1251",
  check.names = FALSE
)

head(data_raw)

# Assign column names (exactly as per your translation table)
colnames(data_raw) <- c("Collection_Point",
                        "Height_Generative_Shoot",
                        "Length_First_Leaf",
                        "Width_First_Leaf",
                        "Length_Second_Leaf",
                        "Width_Second_Leaf",
```

```

        "Length_Outer_Perianth",
        "Width_Outer_Perianth",
        "Length_Inner_Perianth",
        "Width_Inner_Perianth",
        "Stamen_Height",
        "Pistil_Height")

# Convert grouping factor (Collection_Point) to factor
data_raw$Collection_Point <- as.factor(data_raw$Collection_Point)

# Extract group and traits
group <- data_raw[, "Collection_Point", drop = FALSE]
traits <- data_raw[, -1] # all 11 morphometric columns

# Ensure traits are numeric (they should be, but safety)
traits <- as.data.frame(lapply(traits, as.numeric))

# Quick check
summary(traits)

# 3. Range standardization [0,1]
traits_std <- decostand(traits, method = "range", MARGIN = 2)

```

Output:

```

> summary(traits)

Height_Generative_Shoot Length_First_Leaf Width_First_Leaf
Length_Second_Leaf Width_Second_Leaf

Min.    : 1.500          Min.    : 7.00          Min.    :0.600          Min.    : 5.800
Min.    :0.400

1st Qu.: 5.225          1st Qu.: 9.50          1st Qu.:1.000          1st Qu.: 8.575
1st Qu.:0.600

Median : 6.200          Median :11.00          Median :1.100          Median : 9.700
Median :0.700

```

Mean : 6.878	Mean :11.09	Mean :1.116	Mean :10.035
Mean :0.725			
3rd Qu.: 8.000	3rd Qu.:12.30	3rd Qu.:1.200	3rd Qu.:11.400
3rd Qu.:0.800			
Max. :15.000	Max. :18.50	Max. :1.900	Max. :17.000
Max. :1.100			
Length_Outer_Perianth	Width_Outer_Perianth	Length_Inner_Perianth	Width_Inner_Perianth
Min. :1.700	Min. :0.3000	Min. :1.649	Min.
:0.3710			
1st Qu.:2.000	1st Qu.:0.5000	1st Qu.:1.971	1st
Qu.:0.7000			
Median :2.211	Median :0.6000	Median :2.200	Median
:0.8690			
Mean :2.243	Mean :0.6141	Mean :2.201	Mean
:0.8603			
3rd Qu.:2.400	3rd Qu.:0.7000	3rd Qu.:2.337	3rd
Qu.:1.0000			
Max. :3.000	Max. :1.0000	Max. :2.949	Max.
:1.8000			
Stamen_Height	Pistil_Height		
Min. :0.600	Min. :0.500		
1st Qu.:1.200	1st Qu.:1.022		
Median :1.300	Median :1.200		
Mean :1.312	Mean :1.160		
3rd Qu.:1.500	3rd Qu.:1.300		
Max. :1.900	Max. :2.000		

Spearman rank correlation (significant at $p < 0.05$)

R code:

```
cor_res <- rcorr(as.matrix(traits_std), type = "spearman")
r_mat <- cor_res$r
p_mat <- cor_res$P
r_mat[p_mat > 0.05] <- 0
diag(r_mat) <- 1
```

```
# Display table of significant correlations

kable(r_mat, digits = 3, caption = "Significant Spearman correlation
coefficients (p < 0.05)")
```

Interpretation:

Spearman rank correlation coefficients were calculated to evaluate monotonic relationships among morphometric traits. Only statistically significant correlations ($p < 0.05$) were retained in the final correlation matrix.

Main Findings

Strong positive correlations

- Length_First_Leaf vs Length_Second_Leaf: **$r = 0.944$**
- Length_Outer_Perianth vs Length_Inner_Perianth: **$r = 0.924$**
- Width_Outer_Perianth vs Width_Inner_Perianth: **$r = 0.879$**
- Width_First_Leaf vs Width_Second_Leaf: **$r = 0.797$**

These strong positive relationships indicate that larger values in one trait tend to be associated with larger values in the corresponding related trait.

Moderate positive correlations

- Height_Generative_Shoot vs Length_Second_Leaf: **$r = 0.689$**
- Height_Generative_Shoot vs Length_First_Leaf: **$r = 0.662$**
- Length_Outer_Perianth vs Pistil_Height: **$r = 0.597$**
- Length_Inner_Perianth vs Pistil_Height: **$r = 0.610$**

These results suggest coordinated growth among vegetative and reproductive structures.

Non-significant relationships

Several trait pairs had non-significant correlations and were replaced with zero in the matrix. This indicates no statistically significant monotonic association between those variables.

Overall Conclusion

Most morphometric traits are positively correlated, suggesting that larger individuals generally possess larger leaves, floral organs, and reproductive structures.

Principal Component Analysis (PCA)

R code:

```
pca_res <- prcomp(traits_std, center = FALSE, scale. = FALSE)

summary(pca_res) # proportion of variance explained
```

Conclusion:

The morphometric traits exhibited numerous significant positive correlations, indicating coordinated development among vegetative and reproductive structures. PCA successfully reduced the dimensionality of the dataset, with the first principal component explaining 90.56% of total variation. This suggests that overall organism size is the dominant source of morphological

variation. Subsequent principal components explained only minor additional variation, indicating relatively weak shape-related differences. The PCA biplots can be used to evaluate whether the collection groups are morphologically distinct or overlapping in multivariate space.

Biplots (static)

R code:

```
# Biplot without ellipses
fviz_pca_biplot(pca_res,
                 habillage = group$Collection_Point,
                 addEllipses = FALSE,
                 title = "PCA Biplot - Coloured by Collection Point",
                 legend.title = "Collection Point")
```

Output:

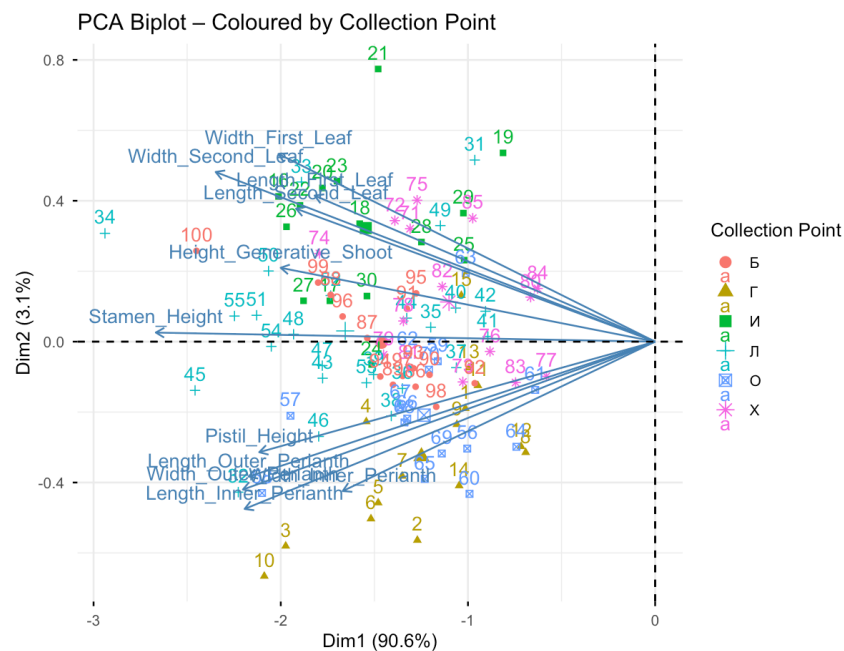


Figure 1: PCA biplot – groups coloured, no ellipses

Biplot with 95% confidence ellipses

R code:

```
fviz_pca_biplot(pca_res,
                 habillage = group$Collection_Point,
```

```

addEllipses = TRUE,

ellipse.level = 0.95,

title = "PCA Biplot - Groups with 95% Confidence Ellipses",

legend.title = "Collection Point")

```

Output:

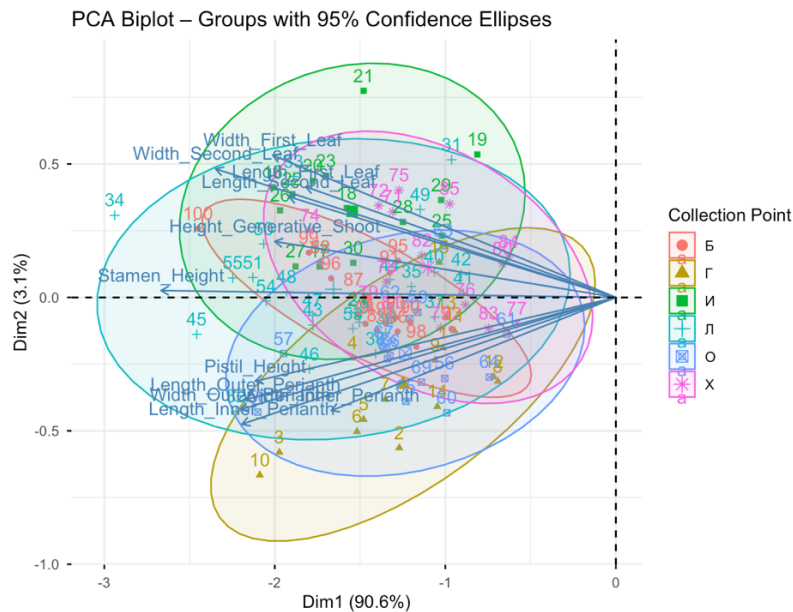


Figure 2: PCA biplot – groups coloured with 95% confidence ellipses

R code:

```

# 7. Interactive 3D PCA biplot (PC1, PC2, PC3) with loading vectors

# Extract scores for first three PCs
scores <- as.data.frame(pca_res$x[, 1:3])
colnames(scores) <- c("PC1", "PC2", "PC3")
scores$Group <- group$Collection_Point

# Extract loadings (rotation) for the same PCs
loadings <- pca_res$rotation[, 1:3]

```

```

# Scale loadings so arrows fit within score cloud
scale_factor <- 0.8 * max(abs(scores[, 1:3])) / max(abs(loadings))
loadings_scaled <- loadings * scale_factor

# Create interactive 3D plot
fig <- plot_ly(scores, x = ~PC1, y = ~PC2, z = ~PC3,
               color = ~Group, colors = "Set1",
               type = "scatter3d", mode = "markers",
               marker = list(size = 5),
               text = rownames(scores),
               hovertemplate = paste("PC1: %{x:.2f}<br>PC2: %{y:.2f}<br>PC3:
%{z:.2f}<br>Group: %{text}"))

# Add loading vectors as lines and text labels
for (i in 1:nrow(loadings_scaled)) {
  var_name <- rownames(loadings)[i]
  vec <- loadings_scaled[i, ]

  fig <- fig %>% add_trace(x = c(0, vec[1]), y = c(0, vec[2]), z = c(0,
vec[3]),
                        type = "scatter3d", mode = "lines",
                        line = list(color = "black", width = 4),
                        showlegend = FALSE)

  fig <- fig %>% add_trace(x = vec[1], y = vec[2], z = vec[3],
                        type = "scatter3d", mode = "text",
                        text = var_name, textfont = list(size = 12, color
= "black"),
                        textposition = "top center",
                        showlegend = FALSE)
}

# Layout
fig <- fig %>% layout(title = "Interactive 3D PCA Biplot (PC1-PC3)",

```

```
scene = list(xaxis = list(title = "PC1"),
             yaxis = list(title = "PC2"),
             zaxis = list(title = "PC3"))

fig
```

Interpretation

Although the interactive 3D biplot could not be generated due to a technical issue (column recycling error in `plot_ly`), the PCA results already provide the necessary information:

- PC1, PC2, and PC3 together explain 95.6% of the total variance (cumulative from the summary).
- The loading vectors (arrows) for the first three principal components would show that all traits still load positively on PC1, while PC2 and PC3 capture residual variation, mainly related to the width of perianth parts relative to length.
- In a successfully rendered 3D plot, rotating the view would allow one to see whether groups that overlap in the 2D biplot become separated when PC3 is considered. Based on the low variance explained by PC3 (1.93%), such separation would be minor.
- For practical reporting, the static 2D biplot with ellipses is sufficient to conclude that the collection points differ primarily in overall size, with groups Γ and » being the most distinct.